O S O S

G T G T

5 E 5 E

N O N O

O S O S

G T G T

5 E 5 E

N O N O

O S O S

G T G T

5 E 5 E

Sequential Decision Making

DONE

1) $\theta_{n+1} = \theta_n + \frac{1}{n}[R_n - \theta_n]$

New Estimate = Old estimate + α_n[Target set − old estimate].

2) $2 = 0 + \alpha_n[4 - 0]$

$\alpha_n = \frac{1}{2}$

3) $0.9 = 0.4 + \alpha_n[4 - 0.4]$

$\frac{0.5}{3.6} \approx \frac{1}{8}$, $\alpha_n = \frac{1}{8}$

4) $4 = 0 + \alpha_n[4 - 0]$

$\alpha_n = 1$

MDPs

Done

4) $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$

$G_0 = \gamma^0 R_1 + \gamma^1 R_2 + \gamma^2 R_3$

$= R_1 + \gamma R_2 + \gamma^2 R_3 = \frac{1}{1-\gamma}$

6) $R_1 = -3$ $R_2 = 5$ $R_3 = 2$ $R_4 = 7$

$R_5 = 1$, $\gamma = 0.8$

$G_0 = -3 + 0.8(5) + 0.8^2 . 2 + 0.8^3 . 7$

$+ 0.8^4 . 1$

$= \underline{\underline{6.27}}$

5, 10, 10, 10, 10

$G_0 = 5 + \gamma G_1$

$G_1 = \frac{10}{1-\gamma} = \frac{10}{1-0.8} = 50$

$\therefore = 5 + 0.8 \times 50 = 45$

7)

$$V_* = \max_a q_*(s,a)$$

8) $$q_* = \sum_{s',r} p(s',r|s,a) \cdot [r + \gamma v_*(s')]$$

9) $\pi_* = 1$ if $a = \arg\max_{a'} q_*(s,a')$ else 0.

10) $\pi_* = 1$ if $v_* = \sum_{s',r} p(s',r|s,a) \cdot [r + \gamma v_*(s')]$

5, 10, 15

$$q_{\pi_{left}}(s, left) = E[G_t | S_t = s]$$

$$= 1 + 0.0$$

$$= 1$$

$$q_{\pi_{right}}(s, right) = 0 + 0.2$$

$$= 0$$

Value functions

DP

$\gamma = 1$

$A \in \{$ Right, Up, Left, Down $\}$.

$q_\pi(11, \text{down})$

$$q_\pi(s,a) = \sum_{s',r} p(s',r \mid s,a) \cdot [r + \gamma \cdot v_\pi(s')]$$

$$= 1[-1 + 1 \cdot 0]$$

$$= -1$$

10

$$V_\pi(s) = \sum_a \pi(a \mid s) \sum_{s',r} p(s',r \mid s,a) \cdot [r + \gamma v_\pi(s')]$$

$$= \pi(\text{left} \mid 15)\{1 \cdot [-1 + x]\}$$

$$+$$

$$\pi(\text{right} \mid 15)\{1 \cdot [-1 + x]\}$$

$$+$$

$$\pi(\text{up} \mid 15)\{1 \cdot \{-1 + 1 \cdot -20\}$$

$$+$$

$$\pi(\text{down} \mid 15)\{1 \cdot \{-1 + x]\}$$

$$= 0.25(-1+x + 0.25(-1+x + 0.25 \cdot (-1+x)$$

$$+ 0.25 \cdot (-21)$$

$$= 0.75(-1+x) + 0.25(-21)$$

$$V(1s) = 0.75(-1+x) + 0.25(-21).$$

$$V_\pi(1s) = -0.75 + 0.25x - 5.25$$

$$V_\pi(1s) = -0.75 + 0.25V_\pi(1s) - 5.25.$$

$$V_\pi(1s) - 0.75V_\pi(1s) = -6$$

$$0.25 \cdot V_\pi(1s) = -6$$

$$V_\pi(1s) = -6 \times 4$$

$$= \underline{\underline{-24}}$$

Policy Improvement theorem

Key → if $q_\pi(s, \pi'(s)) \geq v_\pi(s)$, if this is true then we can say that every time our agent encounters the particular state 's' it will choose action 'a' through the new policy.

Then $v_{\pi'}(s) \geq v_\pi(s)$ and strictly greater where the policy is being changed.

GPI → continuous interaction b/w policy evaluation by calculating v_π and, finding a new policy π' which is greedy w.r.t v_π

$\pi' \sim \text{greedy}(\pi)$, and this goes until

- the policy is greedy w.r.t its own value function, hence converging to an optimal value function with an optimal policy.

$$q_\pi(s,a) = E[G_t \mid S_t = s, A_t = a$$

$$= E[R_{t+1} + \gamma G_{t+1} \mid S_t = s, A_t = a]$$

$$= E[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s, A_t = a]$$

$$= \sum_{s',r} p(s',r \mid s,a) \cdot [r + \gamma v_\pi(s')]$$

$$= \sum_{s',r} p(s',r \mid s,a) \left[r + \gamma \sum_{a'} \pi(a' \mid s') q_\pi(s',a') \right]$$